

Robust Multimodal Fusion for Skin Lesion Classification: A Study of Missing-Modality Tolerance

Nafiur Rahman Afnan
Dept. of Computer Science
BRAC University
Dhaka, Bangladesh
nafiur.rahman.afnan@g.bracu.ac.bd

Nehal Mahfuz
Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
nehal.mahfuz@g.bracu.ac.bd

Tanvirul Hoque
Dept. of Computer Science
BRAC University
Dhaka, Bangladesh
tanvirul.hoque@g.bracu.ac.bd

Abstract—Multimodal skin lesion classifiers combining dermoscopic images and patient metadata achieve high accuracy on benchmarks but degrade when metadata is missing at inference, a frequent occurrence in real clinics. We study *Modality Dropout*, a simple training-time augmentation that zeros the metadata vector with probability p , on two dermatology datasets. On HAM10000, the baseline already ignores metadata due to image dominance (image-only F1 = 0.720, metadata-only F1 = 0.242), leaving no robustness problem to solve. We therefore evaluate on PAD-UFES-20, where metadata is genuinely informative (metadata-only F1 = 0.609, exceeding image-only at 0.558) and 35% of records contain partial missingness. Across five dropout levels and three random seeds, no setting beats the baseline on macro F1; higher p monotonically reduces both peak performance and decay. However, per-class analysis reveals that Modality Dropout substantially stabilizes recall on majority classes: ACK loses only 1.5 points versus baseline’s 19.7 points under full metadata removal, and the method perfectly preserves melanoma detection, at the cost of degraded minority-class performance. Aggregate metrics obscure this clinically meaningful trade-off.

Index Terms—multimodal learning, missing modality, modality dropout, skin lesion classification, HAM10000, PAD-UFES-20

GitHub: https://github.com/tanvir75/skin_lesion_classification_modarity_dropout_evaluation

I. INTRODUCTION

Skin cancer diagnosis from dermoscopic and clinical images is a well-studied vision task. Multimodal models combining images with patient metadata such as age, body site, symptoms, and lifestyle factors consistently improve over image-only baselines [4], [6]. Yet clinical deployment exposes a fragility: in busy clinics, metadata is often missing. Patients may not know certain answers, or staff lack time to record them. A model trained on complete data may degrade unpredictably when its metadata input is absent.

Modality Dropout offers a simple, architecture-agnostic remedy: during training, randomly replace the metadata vector with zeros with probability p . The model implicitly learns to handle both conditions. We empirically test this method on skin lesion classification, with results that motivate a more nuanced view of when it helps.

We initially studied **HAM10000** [1] with 10,015 dermoscopic images across seven classes. Our baseline achieved 0.749 test macro F1 but showed near-zero decay under missing metadata. A diagnostic decomposition revealed why: image-only performance (0.720 F1) was nearly identical to multimodal (0.749 F1), and metadata-only was only 0.242 F1, barely above random. The model ignored the weak metadata signal, and there was no robustness problem to study.

We pivoted to **PAD-UFES-20** [7], a smaller dataset of 2,298 smartphone images across six classes with 22 clinical metadata fields. Here metadata is genuinely diagnostic: metadata-only F1 (0.609) exceeds image-only (0.558), and 35% of records already contain real missing data, making the problem authentic rather than synthetic. We trained baseline and Modality Dropout models at $p \in \{0.10, 0.15, 0.30, 0.50\}$, three seeds each, and evaluated under four missing rates.

Our contributions are:

- 1) A cross-dataset diagnostic showing the missing-modality robustness problem itself depends on dataset characteristics.
- 2) A five-level Modality Dropout ablation on PAD-UFES-20 demonstrating a monotonic peak-versus-stability trade-off with no sweet spot superior to the baseline in macro F1.
- 3) Per-class evidence that Modality Dropout meaningfully stabilizes majority-class and melanoma recall under missing metadata, a finding hidden by macro F1.

II. RELATED WORK

Multimodal skin lesion classification. Tschandl *et al.* [1] introduced HAM10000, the standard dermoscopic benchmark. FusionM4Net [4] proposes hierarchical two-stage multimodal fusion achieving 77% accuracy on the Seven-Point Checklist but assumes complete metadata at inference. SM3 [6] uses self-supervised contrastive learning between paired dermoscopic and clinical images, combined with clustered pseudo-multi-labels representing seven lesion attributes; it does not address missing modalities. SCGViT [5] uses a lightweight ViT with synthesized pseudo-multimodal branches generated

by a conditional GAN, but relies on synthetic samples rather than real patient metadata.

Missing-modality robustness. ShaSpec [2] is the closest related work, proposing shared-specific feature extraction with auxiliary tasks for missing modalities in medical segmentation, achieving 3 to 5% gains on BraTS. It is not applied to skin lesion classification with tabular metadata.

Evaluation rigor and datasets. Codella *et al.* [3] introduced stricter ISIC evaluation metrics and showed top models fail on >10% of images, motivating our per-class analysis. Pacheco *et al.* [7] released PAD-UFES-20, the smartphone dermatology dataset we adopt.

To our knowledge, no prior work combines (i) skin lesion classification, (ii) rich tabular clinical metadata, and (iii) systematic missing-modality evaluation. This work fills that gap.

III. METHODOLOGY

A. Datasets

We use two skin lesion datasets with contrasting characteristics, summarized in Table I.

TABLE I: Dataset comparison. PAD-UFES-20 is smaller but has substantially richer metadata and a much higher rate of real-world missingness in the original records.

Property	HAM10000	PAD-UFES-20
Images	10,015	2,298
Patients	not provided	1,373
Classes	7	6
Image type	dermoscopic	smartphone
Metadata fields	3	22
Metadata dim	19	60
Real missingness	<1%	35%
Largest class	NV (67%)	BCC (37%)
Smallest class	VASC (<1.5%)	MEL (2.3%)

HAM10000 [1] contains 10,015 dermoscopic images across seven diagnostic classes: melanocytic nevi (NV, 67%), melanoma (MEL), basal cell carcinoma (BCC), actinic keratosis (AKIEC), benign keratosis (BKL), dermatofibroma (DF), and vascular lesions (VASC). Per-patient metadata is limited to age, sex, and lesion localization (15 body sites). Age is missing for 57 records; we impute the median.

PAD-UFES-20 [7] contains 2,298 smartphone-acquired clinical images from 1,373 patients across six classes: BCC (37%), actinic keratosis (ACK, 32%), nevus (NEV), seborrheic keratosis (SEK), squamous cell carcinoma (SCC), and melanoma (MEL, 2.3%). The dataset includes 22 clinical metadata fields covering lifestyle (smoking, alcohol, pesticide exposure), family history (skin and other cancer), lesion symptoms (itching, growth, bleeding, change, hurt, elevation), and demographics (age, gender, Fitzpatrick skin type, body region, lesion diameter). Importantly, **35% of records have substantial missing metadata** in the original data, reflecting pediatric patients, partial records, and unfilled questionnaires.

Multimodal Skin Lesion Classification Pipeline

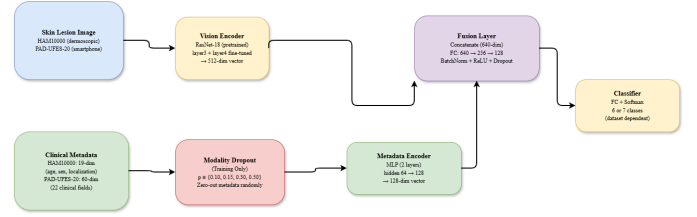


Fig. 1: Three-component multimodal architecture. Modality Dropout is applied at training time only, inside the dataset class, allowing independent control of the missing rate at inference. The same architecture is used for both HAM10000 (METADIM = 19) and PAD-UFES-20 (METADIM = 60).

This makes PAD-UFES-20’s missing-modality scenario clinically authentic rather than purely synthetic. For PAD-UFES-20, we split at the *patient level* (70/10/20 train/val/test) so no patient appears in both training and testing.

B. Metadata Encoding

For HAM10000, the metadata vector has dimension 19: 1 normalized age, 3 one-hot sex categories, and 15 one-hot body sites. For PAD-UFES-20, the vector has dimension 60: 4 normalized numerical features (age, two lesion diameters, Fitzpatrick type), 39 binary one-hot features (13 yes/no fields each with TRUE/FALSE/UNK indicators), 3 gender categories, and 14 body regions. Missing numerical values are imputed with the median; missing binary or categorical values are encoded as a dedicated “UNK” or “unknown” category, preserving the missingness signal.

C. Model Architecture

Our model has three components (Fig. 1):

- 1) **Vision Encoder.** A ResNet-18 [8] pretrained on ImageNet. The early stages (conv1, layer1, layer2) are frozen; layer3 and layer4 are fine-tuned. Global average pooling produces a 512-dimensional image feature vector.
- 2) **Metadata Encoder.** A two-layer MLP with hidden size 64 and output size 128, with ReLU activations and dropout (0.3) between layers. Input dimension matches the dataset’s metadata vector (19 for HAM10000, 60 for PAD-UFES-20).
- 3) **Fusion Classifier.** Image and metadata features are concatenated (640 dimensions), then passed through two fully-connected layers (256 → 128) with batch normalization, ReLU, and dropout (0.4), followed by a linear classifier producing six or seven class logits.

We chose ResNet-18 over larger backbones to keep the experimental contribution focused on the training paradigm rather than encoder capacity, and to ensure the same fusion architecture across all experiments.

D. Modality Dropout

During training, with probability p , the entire metadata vector for a given training example is replaced with zeros. The model is not informed which examples are dropped; it must implicitly learn to handle both conditions. We implement Modality Dropout inside the `Dataset.__getitem__` method rather than in the model’s forward pass. This cleanly decouples the training-time augmentation from the model architecture and allows independent control of the missing rate at test time.

At inference, we evaluate models at fixed missing rates by setting the dataset’s dropout probability to the desired test value (0.00, 0.25, 0.50, 1.00). This simulates a controlled clinical scenario where a known fraction of patient questionnaires arrives incomplete.

E. Training Setup

We train with the Adam optimizer (learning rate 10^{-4} , weight decay 10^{-5}), weighted cross-entropy loss with class weights computed via `sklearn`’s balanced strategy, batch size 32, and a `ReduceLROnPlateau` scheduler (patience 5, factor 0.5) monitoring validation macro F1. We train for up to 50 epochs with early stopping (patience 10). Images are resized to 224×224 , with random horizontal/vertical flips, $\pm 20^\circ$ rotation, and color jitter for training augmentation, followed by ImageNet normalization.

For baseline runs, the training dropout probability is 0 and the validation set is evaluated with full metadata. For Modality Dropout runs, the validation set is evaluated at the same dropout rate used during training. This aligns the checkpoint-selection metric with the model’s deployment distribution and prevents premature early stopping observed in pilot experiments.

F. Evaluation Protocol

We evaluate each trained model at four missing-metadata rates (0%, 25%, 50%, 100%) and report **macro F1**, **accuracy**, and **per-class recall**. Macro F1 is our primary metric due to severe class imbalance, since accuracy can be misleading when one class dominates. For statistical reliability, all main experiments are repeated across three random seeds (42, 123, 2024); we report mean $\pm$ standard deviation.

We also conduct two *diagnostic* experiments per dataset to characterize where the diagnostic signal lives: a metadata-only model (just the metadata encoder and classifier) and an image-only model (just the vision encoder and classifier). The gap between these and the multimodal model quantifies the value of fusion and predicts whether missing-modality robustness can be a meaningful problem on that dataset.

IV. EXPERIMENTS AND RESULTS

We organize our findings around three questions: (1) Does HAM10000 support the missing-modality problem we set out to study? (2) Does PAD-UFES-20 support it? (3) Does Modality Dropout help on a dataset where the problem is real?

A. The HAM10000 Experiment

We first trained our baseline multimodal model on HAM10000 for 46 epochs (early stopping triggered), reaching a best validation macro F1 of 0.781. On the test set, the model achieved 0.749 macro F1 with full metadata, a strong result by HAM10000 standards.

We then tested the same model at four missing-metadata rates. The result was unexpected (Table II): the model showed *no decay*. Macro F1 was effectively constant whether we provided full metadata or zeroed it out completely. This contradicted our hypothesis.

TABLE II: Baseline decay on HAM10000 (single seed). The model shows no degradation under missing metadata.

Missing	Macro F1	Accuracy
0%	0.749	0.825
25%	0.748	0.827
50%	0.748	0.830
100%	0.749	0.830

To understand why, we ran two diagnostic experiments: a metadata-only model and an image-only model. The results (Table III, HAM10000 column) explained the puzzle. Metadata alone barely beats random (0.242 F1). The image alone gets 0.720, and adding metadata pushes this up by just 0.029 points. The multimodal model rationally learned that metadata was not worth using; it routed predictions through the image features. When we zeroed the metadata at test time, nothing happened, because the model never used it. **HAM10000’s metadata is too weak to support a missing-modality robustness study.**

This is itself an informative finding: it suggests that benchmarks with thin metadata may give the false impression that multimodal models are implicitly robust, when in fact they are simply ignoring one modality.

B. Pivot to PAD-UFES-20

We applied the same diagnostic decomposition to PAD-UFES-20 (Table III, PAD-UFES-20 column). The signal structure is inverted relative to HAM10000: **metadata alone (0.609) actually outperforms image alone (0.558)**. This makes clinical sense, since questions like “has the lesion grown? does it bleed? is there a family history of cancer?” are the same ones a dermatologist asks during a real exam. The multimodal model gains 0.018 macro F1 over metadata-only, reflecting modest but real fusion benefit. Fig. 2 visualizes this contrast.

TABLE III: Cross-dataset modality decomposition. Test macro F1. HAM10000 is image-dominant; PAD-UFES-20 is metadata-dominant.

Model	HAM10000	PAD-UFES-20
Random	≈ 0.143	≈ 0.167
Metadata only	0.242	0.609
Image only	0.720	0.558
Multimodal	0.749	0.627

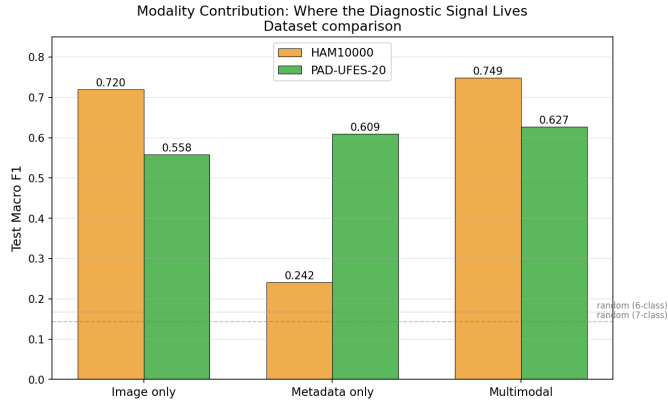


Fig. 2: Cross-dataset modality contribution. On HAM10000, images dominate; on PAD-UFES-20, metadata dominates. The latter is where missing-modality robustness becomes a meaningful research question.

C. Modality Dropout Ablation

We trained models at five Modality Dropout levels: $p = 0.00$ (baseline), 0.10, 0.15, 0.30, and 0.50. The first four were trained with three random seeds (42, 123, 2024); $p = 0.50$ used a single seed as an ablation upper bound. Each model was evaluated at four missing-metadata rates on the test set. Table IV reports macro F1 with standard deviations across seeds, and Fig. 3 visualizes the decay curves.

Three patterns are immediately visible:

- 1) **Higher dropout reduces decay.** Decay shrinks monotonically from +0.047 (baseline) to +0.003 ($p=0.50$). The robust models are flatter across missing rates.
- 2) **Higher dropout reduces peak performance.** Macro F1 at 0% missing also drops monotonically: $0.627 \rightarrow 0.603 \rightarrow 0.597 \rightarrow 0.583 \rightarrow 0.542$.
- 3) **The baseline beats every robust setting at every missing rate tested.** No (p , missing rate) combination achieves higher macro F1 than the baseline.

Fig. 4 plots peak macro F1 against decay magnitude for each dropout level. The points form a clean trade-off curve, but the baseline dominates: there is no Pareto-superior dropout setting in macro F1 terms.

D. Per-Class Stability Analysis

Macro F1 averages performance across all six classes equally, which obscures per-class effects. We compute per-class recall and the recall drop induced by removing all metadata (Table V). Fig. 6 visualizes the stability gain per class.

The picture changes substantially from macro F1. Looking at recall drops:

- **ACK** (the second-largest class): baseline loses 19.7 recall points when metadata is removed; the robust model loses only 1.5, a large stability gain.
- **BCC** (the largest class): baseline loses 12.7 points; robust loses 5.4.

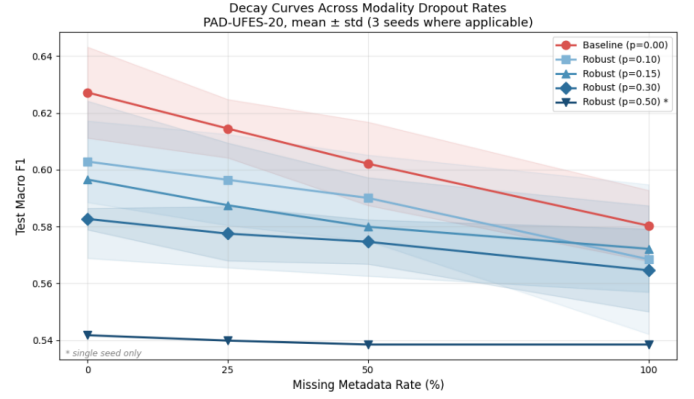


Fig. 3: Decay curves across Modality Dropout rates on PAD-UFES-20. Mean $\pm$ std across 3 seeds (single seed for $p = 0.50$). Higher dropout flattens the curve but lowers its overall level.

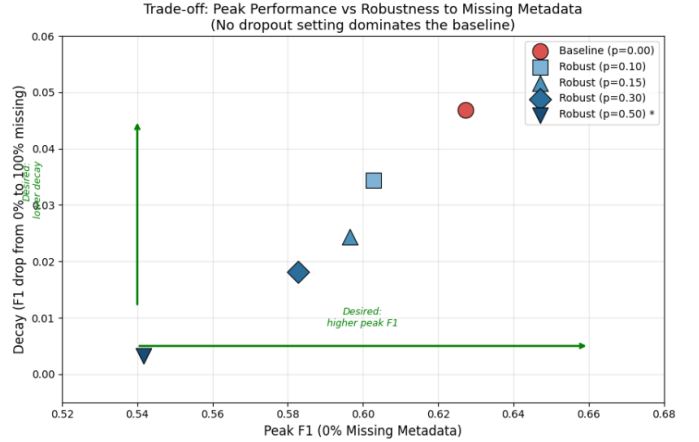


Fig. 4: Trade-off between peak F1 (at 0% missing) and decay (drop from 0% to 100% missing). The baseline ($p = 0.00$) sits in the dominated quadrant, but no robust setting Pareto-dominates it: every reduction in decay is paid for with a larger reduction in peak.

- **MEL** (clinically critical): baseline loses 6.7 points; **robust loses 0**, a perfect preservation of melanoma detection under full metadata removal.
- **SCC** (minority class): the robust model collapses from 0.375 to 0.150 even with full metadata; the regularization pushes SCC predictions toward the visually similar BCC class.
- **NEV, SEK**: small effects.

So while the robust model loses on macro F1, it wins on three of the six classes, including the two largest (BCC, ACK) and the clinically most important (MEL). The macro F1 loss is driven by the SCC collapse and a small loss on SEK. **Modality Dropout produces a clear class-dependent trade-off that aggregate metrics do not reveal.**

TABLE IV: Macro F1 across dropout levels and missing-metadata rates (mean $\pm$ std, 3 seeds where applicable). The baseline outperforms every robust setting at every missing rate.

Model	0% miss	25% miss	50% miss	100% miss	Decay
Baseline ($p=0.00$)	0.627 $\pm$ 0.016	0.614 $\pm$ 0.010	0.602 $\pm$ 0.015	0.580 $\pm$ 0.012	+0.047
Robust ($p=0.10$)	0.603 $\pm$ 0.014	0.596 $\pm$ 0.016	0.590 $\pm$ 0.015	0.568 $\pm$ 0.026	+0.034
Robust ($p=0.15$)	0.597 $\pm$ 0.028	0.588 $\pm$ 0.022	0.580 $\pm$ 0.017	0.572 $\pm$ 0.015	+0.024
Robust ($p=0.30$)	0.583 $\pm$ 0.004	0.578 $\pm$ 0.010	0.575 $\pm$ 0.008	0.565 $\pm$ 0.015	+0.018
Robust ($p=0.50$)*	0.542	0.540	0.539	0.539	+0.003

* Single seed only.

Confusion Matrices: Baseline vs Robust at 0% and 100% Missing Metadata (numbers in parentheses are raw counts; seed=42)

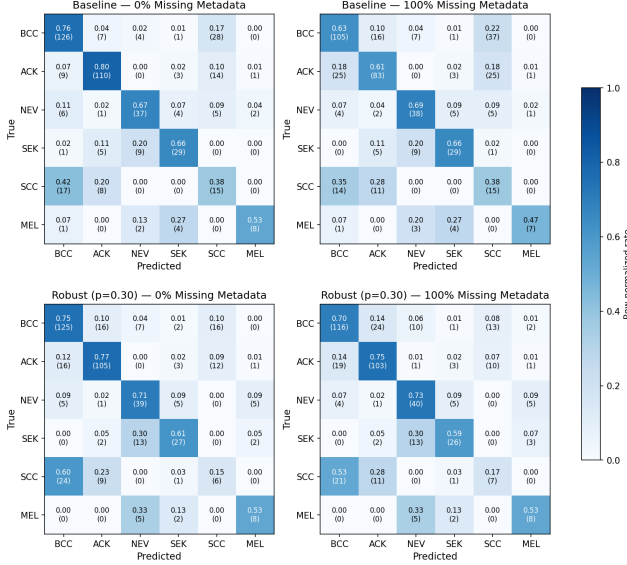


Fig. 5: Confusion matrices for baseline and robust ($p=0.30$) models at 0% and 100% missing metadata (seed = 42). Cells are row-normalized; raw counts in parentheses. The bottom-right panel shows that under full metadata removal, the robust model reassigns most SCC cases to BCC (21/40), explaining the SCC collapse seen in Table V. Both models also confuse SEK with NEV, and MEL is distributed across NEV and SEK.

TABLE V: Per-class recall at 0% and 100% missing metadata (seed = 42). The robust model preserves majority-class and MEL recall, but degrades SCC.

Class	n	B 0%	B 100%	R 0%	R 100%
BCC	166	0.759	0.633	0.753	0.699
ACK	137	0.803	0.606	0.766	0.752
NEV	55	0.673	0.691	0.709	0.727
SEK	44	0.659	0.659	0.614	0.591
SCC	40	0.375	0.375	0.150	0.175
MEL	15	0.533	0.467	0.533	0.533

B = Baseline, R = Robust ($p=0.30$).

V. DISCUSSION

A. When Does the Missing-Modality Problem Exist?

Our cross-dataset comparison reveals that the missing-modality robustness problem is not universal; it depends

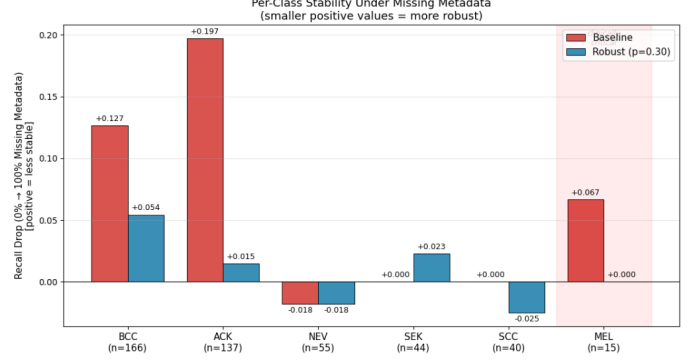


Fig. 6: Per-class recall drop when going from 0% to 100% missing metadata. Smaller positive values indicate more stable behavior. Modality Dropout substantially stabilizes ACK (19.7 points $\rightarrow$ 1.5), BCC (12.7 $\rightarrow$ 5.4), and perfectly preserves MEL (6.7 $\rightarrow$ 0), at the cost of degraded SCC and SEK.

on whether the multimodal model actually relies on both modalities. On HAM10000, image-only F1 (0.720) was nearly identical to multimodal (0.749). The model rationally ignored the weak metadata signal, so removing it at test time caused no measurable degradation. This is not robustness; it is implicit single-modality behavior masquerading as robustness.

On PAD-UFES-20, metadata alone (0.609 F1) outperforms image alone (0.558). The multimodal model genuinely uses both, and removing metadata causes measurable decay (0.047 macro F1 points for the baseline). This is the kind of dataset where missing-modality research can meaningfully be conducted.

Our diagnostic methodology, training simple metadata-only and image-only baselines before launching a robustness study, is a cheap and informative check we recommend for any future multimodal missing-modality research. A small gap between image-only and multimodal performance is a warning sign that any “robustness” demonstrated may be misleading.

B. Why Modality Dropout Does Not Improve Macro F1

Modality Dropout failed to improve macro F1 on PAD-UFES-20 at any tested dropout level. We attribute this to two factors.

First, **PAD-UFES-20 already contains 35% real missing metadata** in the form of “UNK” or “unknown” indicators. The baseline model, trained on this data, already learns to handle missingness implicitly; its 0.047 F1 gap between 0%

and 100% missing reflects partial baseline robustness. Adding aggressive synthetic Modality Dropout ($p=0.30$ or higher) on top of this is over-regularization: the model effectively sees missing metadata on more than half of training batches, and learns to weight metadata less.

Second, **the regularization pushes predictions toward the majority classes.** This shows clearly in Table V and Fig. 5: SCC recall drops from 0.375 (baseline) to 0.150 (robust at $p = 0.30$), with the lost SCC predictions reassigned to the visually similar BCC class. The minority class collapse pulls macro F1 down enough to outweigh the per-class gains on majority classes.

C. Per-Class Trade-Off and Clinical Implications

The most important finding of this work is hidden by macro F1. When we examine per-class recall stability under missing metadata (Fig. 6), Modality Dropout produces clear benefits on three classes and clear costs on one.

The melanoma finding is clinically meaningful. Melanoma is the most dangerous skin cancer and the diagnosis where AI assistance is most valuable. The fact that Modality Dropout perfectly preserves melanoma recall under full metadata absence, while the baseline loses one of its eight detected melanoma cases, is exactly the kind of behavior a deployed clinical model should exhibit. A clinician forgetting to fill out the patient questionnaire should not cost a patient their melanoma diagnosis.

Macro F1, by weighing all six classes equally, sees the SCC collapse as a four-times-larger contribution to the average than the MEL gain (because SCC has 40 test cases versus MEL's 15). This penalizes Modality Dropout disproportionately to its clinical cost. A reasonable clinical metric weighted by case severity would likely favor the robust model.

D. Limitations

Three limitations qualify our findings. (1) PAD-UFES-20 is a relatively small dataset (1,597 training images) with severe class imbalance; MEL has only 15 test cases, making the perfect-preservation finding subject to a wide confidence interval. (2) We did not explore architectural changes beyond ResNet-18 with concatenation fusion. More sophisticated fusion (attention-based, gated, dual-branch) might handle the regularization differently. (3) PAD-UFES-20's smartphone images have lower visual quality than HAM10000's dermoscopic images, which limits how well our findings generalize to clinical dermoscopy.

VI. CONCLUSION

We empirically investigated Modality Dropout for missing-modality robustness in multimodal skin lesion classification. Our work began on HAM10000, where we discovered the baseline model already ignored metadata, leaving no robustness problem. We pivoted to PAD-UFES-20, where metadata is genuinely diagnostic and 35% of records contain authentic missing data.

Across five dropout levels and three random seeds, Modality Dropout did not improve macro F1 over the baseline. However, per-class analysis revealed a finding hidden by aggregate metrics: the method substantially stabilizes recall on majority classes (ACK, BCC) and perfectly preserves melanoma detection under full metadata removal, at the cost of degraded performance on the minority class SCC.

We argue that **missing-modality robustness in clinical AI should be evaluated per-class with clinical severity in mind**, not solely with macro F1. We also recommend that any missing-modality study include image-only and metadata-only diagnostic baselines to verify that the dataset under study actually supports the research question.

Future work should explore (i) class-weighted training objectives that prevent minority-class collapse under regularization, (ii) attention-based fusion architectures with explicit missingness signals, and (iii) prospective validation on larger, dermoscopic, real-world multimodal datasets.

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